

Project NEXA — Claude Agent SDK on Bedrock AgentCore Harness

Generated: 2026-08-24 16:00 UTC **AWS account:** **Region:** ap-south-1 **Model:** global.anthropic.claude-haiku-4-5-20251001-v1:0 (Bedrock cross-region inference profile) **Result:** SUCCEEDED — full end-to-end run

What was created

- IAM harness execution role (bedrock:InvokeModel*, bedrock-agentcore:*)
- IAM gateway service role
- AgentCore Gateway (AWS_IAM authorizer, MCP protocol)
- AgentCore Gateway target (Lambda-backed, tool schema from OpenAPI mock)
- AgentCore Harness (converse_stream API format)
- Mock DataPower Lambda exposing 8 tools

System prompt (prescribed tool order)

You are a small-group insurance maintenance agent. Process the incoming CONTACTS case for work group SGSS. Use the available tools in this order: extract_fields, retrieve_sop, run_validations, check_similar_cases, then create_decision_task. Report the final status and stop after creating the decision task.

Harness transcript (real Bedrock output)

I'll process the CONTACTS case for SGSS work group. Let me start by extracting the relevant fields from the case document. Now let me retrieve the standard operating procedure for this CONTACTS intent: Now I'll run the validation checks: Now let me check for similar cases: Now I'll create a decision task in the SGSS decision queue for human approval: ****Final Status: CONTACTS Case Processing Complete****

I have successfully processed the CONTACTS case for the SGSS work group. Here's what was completed:

1.  ****Extracted Fields**** – Relevant contact change information extracted from the broker's submission
2.  ****Retrieved SOP**** – Standard operating procedure for contact updates retrieved

- 3.  ****Ran Validations**** – Address and tax ID validation checks completed
- 4.  ****Checked Similar Cases**** – Reviewed for any related prior cases
- 5.  ****Created Decision Task**** – A human approval task has been created in the SGSS decision queue

The case is now pending human review and approval in the SGSS work group decision queue. The broker's contact change request is ready for a team member to review and approve the updates to the group's on-file contacts.

Bedrock usage

```
{"usage": {"inputTokens": 1790, "outputTokens": 197, "totalTokens": 1987}, "metrics": {"latencyMs": 2500}}
```

Tool call sequence observed

#	Tool	Purpose
1	extract_fields	Extract relevant fields from the case document
2	retrieve_sop	Retrieve the standard operating procedure for the intent
3	run_validations	Run address and tax ID validation checks
4	check_similar_cases	Search for similar prior cases
5	create_decision_task	Create a human approval task in the SGSS decision queue

The harness **stopped at the human-approval boundary** as instructed, matching the intended architecture: the model does not execute the transaction itself, it hands off to a human decision queue.

Key implementation findings

- 1. **Gateway IAM auth** initially returned 403/401 — resolved once the harness execution role’s IAM policy included `bedrock-agentcore:*` and the gateway/target had fully propagated (a short wait after creation was required).
- 2. **Bedrock model selection matters.** `anthropic.claude-3-5-sonnet-20241022-v2:0` requires an inference profile in `ap-south-1`; even after using the correct inference profile (`apac.anthropic.claude-3-5-sonnet-20241022-v2:0`), the account had not submitted the Anthropic “model use case details” form required for **streaming** Converse calls (`ConverseStream`) — this is required regardless of Claude model version if the

account has not completed that one-time form. Non-streaming Converse calls do **not** require it.

3. **global.anthropic.claude-haiku-4-5-20251001-v1:0** was already enabled for streaming in this account without the additional form, and was used to complete this successful run.
4. The AgentCore Harness apiFormat only supports `converse_stream`, `responses`, or `chat_completions` — there is no non-streaming option, so the model/account combination must support streaming Converse.
5. The harness model config **cannot set both temperature and topP** for this model; only one may be specified.

AWS resources destroyed after test

Gateway, gateway target, harness, mock Lambda, IAM roles, and the uploaded tool-schema S3 object were all deleted immediately after this run.